AEROSPACE STRUCTURES LAB

V Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIA	SEE	Total
A6AE21	PCC	-	-	3	1.5	40	60	100

LIST OF EXPERIMENTS

1.Fabrication & Testing of riveted joints.
2. Verification of Maxwell's and Castiglianos theorems
3. Determination of critical load for a column – (i) Both ends hinged; (ii) Both ends fixed.
4. Determination of critical load for a column – One end fixed other end hinged.
5. NDT inspection – Dye penetration, Magnetic particle and Ultrasonic testing
6. Shear centre of an open section beam
7. Shear centre of a closed section beam
8. Determination of natural frequency of beam undergoing free longitudinal vibrations
9. Determination of natural frequency of Shaft undergoing torsional vibrations
10. Determination of natural frequency for a beam – Forced vibration

REFERENCE BOOKS:

1. Megson T. H. G (2012), Aircraft Structures for Engineering Students, 5th edition, Elsevier, New York.
2.Bruhn (1973), Analysis and Design of Flight Vehicle Structures, Tri-state Offset Company, USA
3.B. C. Punmia (2011), Theory of Structures, 13th edition, Laxmi Publication, Hyderabad
4. Timoshenko, Mechanics of Materials, CBS Publication

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Locate the shear centre for open and closed section beams
2. Estimate the crippling load for long column & Verification of Maxwell's theorems and Castiglianos theorems
3. Identify the surface flaws as well as internal flaws using major NDT techniques.
4. Determine the natural frequency under forced and free vibration
5. Exemplify the fabrication and failure of rivets